

Excel for Analytics

The World's Most Used Data Tool

■ EXCEL MASTERY

■ LEARNING OUTCOME

By the end of this PDF, you will use Excel confidently for data analysis: writing formulas, building pivot tables, creating charts, and automating repetitive tasks.

■ Skills You Will Gain

- Write essential Excel formulas for analytics
- Build and customize Pivot Tables
- Create professional charts
- Use VLOOKUP, INDEX-MATCH for lookups
- Apply conditional formatting for insights
- Filter, sort, and group data efficiently

■ 90% of business analytics still happens in Excel. Even Python users need Excel for reports and stakeholder communication.

SECTION 1

Essential Analytics Formulas

Formula	What It Does	Example
=SUM(A2:A100)	Add all values	Total sales
=AVERAGE(B2:B50)	Calculate mean	Average salary
=COUNTIF(C:C,'Yes')	Count matching values	Count approved orders
=SUMIF(B:B,'>5000',C:C)	Sum where condition is true	Sum sales >5000
=MAX(D:D) / =MIN(D:D)	Highest / Lowest value	Best and worst month
=IFERROR(formula, 0)	Handle errors gracefully	Avoid #DIV/0! errors
=TEXT(A2,'DD-MMM-YYYY')	Format dates as text	Show date nicely
=TRIM(A2)	Remove extra spaces	Clean text data

SECTION 2

VLOOKUP & INDEX-MATCH

VLOOKUP — Look up a value in a table

=VLOOKUP(lookup_value, table_range, col_index, FALSE) Example: =VLOOKUP(A2, Products!A:C, 3, FALSE) → Finds product name in A2, returns price from column 3

INDEX-MATCH — More powerful than VLOOKUP

=INDEX(return_range, MATCH(lookup_value, lookup_range, 0)) Example: =INDEX(C2:C100, MATCH(A2, B2:B100, 0)) → Works for left-side lookups; faster on large datasets

SECTION 3

Pivot Tables — Step by Step

1	Select Data Click anywhere in your data table (Ctrl+A to select all)
2	Insert Go to Insert → PivotTable → New Worksheet → OK
3	Rows Drag 'Category' or 'Month' to ROWS area
4	Values Drag 'Sales' to VALUES area (auto-sums by default)

5

Columns

Drag 'Region' to COLUMNS for a cross-tab view

6

Filter

Drag 'Year' to FILTERS to slice by year

Right-click Values → Summarize by (Count, Average, Max) Use PivotChart for instant visual from your pivot table
 Ctrl+Shift+L adds filters instantly to any table

SECTION 4

Charts for Analytics

Chart Type	Best For	How to Create
Bar Chart	Compare categories (Sales by Region)	Select data → Insert → Bar Chart
Line Chart	Show trends over time	Select data → Insert → Line Chart
Pie Chart	Show proportions (<6 categories)	Select data → Insert → Pie
Scatter Plot	Show correlation between two variables	Insert → Scatter
Histogram	Show frequency distribution	Insert → Statistical → Histogram

SECTION 5

Conditional Formatting

• Home → Conditional Formatting → Highlight Cells Rules → Greater Than • Data Bars: Show a mini bar chart inside cells • Color Scales: Red-Yellow-Green heat map for quick scanning • Icon Sets: Arrows up/down for trend indication • Top/Bottom Rules: Highlight top 10 or bottom 10 performers

■ PRACTICE QUESTIONS

Q1. Write a formula to count all orders where Status = 'Delivered'.

Answer: =COUNTIF(B:B, 'Delivered')

Q2. What does VLOOKUP's last argument FALSE mean?

Answer: It means Exact Match — only return a result if it exactly matches the lookup value.

Q3. When would you use INDEX-MATCH over VLOOKUP?

Answer: When you need to look up values to the left of the key column, or for better performance on large files.

Q4. What is a Pivot Table used for?

Answer: Summarizing, grouping, and aggregating large datasets without writing formulas.

Q5. Write a SUMIF formula to sum sales only for the 'North' region.

Answer: =SUMIF(C:C, 'North', D:D) — assuming C=Region, D=Sales

■ MINI PROJECT

Apply what you've learned with a real dataset!

Dataset: sales_data.xlsx (Date, Product, Region, Units, Revenue)

Steps:

1	Clean Remove blanks, format dates correctly
2	Pivot 1 Revenue by Month — identify best month
3	Pivot 2 Units sold by Product — top product
4	Pivot 3 Revenue by Region — geographic comparison
5	Charts Line chart for monthly trend, bar chart for region
6	Format Add conditional formatting to highlight top performers

Expected Output: A 1-page Excel dashboard with 3 charts and summary KPIs

Insights to Find: Top month, top product, top region, month-over-month growth %

■ Excel Analytics Cheat Sheet	
<code>=SUM(range)</code>	Total of all values
<code>=AVERAGE(range)</code>	Mean value
<code>=COUNTIF(range, crit)</code>	Count matching values
<code>=SUMIF(rng, crit, sum_rng)</code>	Sum where condition met
<code>=VLOOKUP(v, tbl, col, 0)</code>	Look up value in table
<code>=IFERROR(formula, 0)</code>	Catch errors, return 0
<code>=TRIM(text)</code>	Remove extra whitespace
<code>=IF(cond, true, false)</code>	Conditional value
<code>Ctrl + T</code>	Convert range to Table
<code>Alt + N + V</code>	Insert Pivot Table
<code>Ctrl + Shift + L</code>	Toggle filters
<code>F11</code>	Create chart from selection